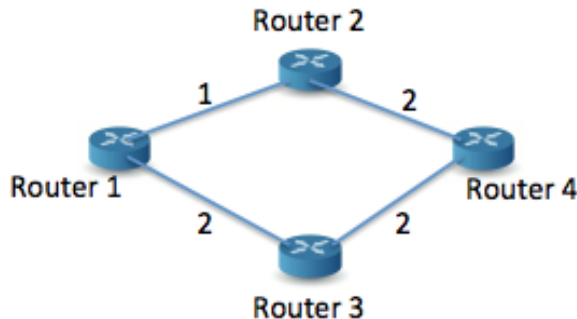


Exercise 1:

Write the content of all Distance Vector messages sent by Router 1 to the other routers in the two following cases: (i) “plain” Distance Vector is used (i.e., without Split Horizon), (ii) Distance Vector is adopted with Split Horizon with Poisonous Reverse and Hop-limit=16. The following figure illustrates the network topology and the routing table inside Router 1.



Destinazione	Next Hop	Costo
Net 1	Router 2	4
Net 2	Router 3	3
Net 3	Router 2	2
Net 4	Router 3	3
Net 5	Local	Local

Solution:

“Plain” Distance Vector (without Split Horizon):

Router 1 sends the same message to all connected (neighbor) routers:

Net 1:4, Net2:3, Net3:2, Net4:3, Net5: local

Distance Vector with Split Horizon:

Router 1 sends different messages to Router 2 and Router 3.

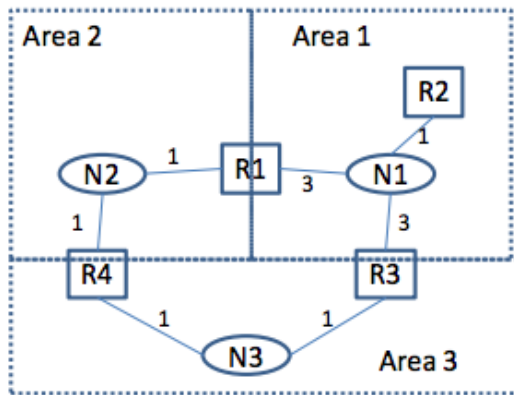
The destinations that Router 1 can reach using as next hop the Router that receives the DV message are indicated with a cost equal to the Hop Limit in all sent DV messages.

To Router 2: Net 1: 16, Net 2: 3, Net 3: 16, Net 4: 3, Net 5: local

To Router 3: Net 1: 4, Net 2: 16, Net 3: 2, Net 4: 16, Net 5: local

Exercise 2

The OSPF protocol is used in the routing domain illustrated in the following figure. Such routing domain is divided into 3 Areas, as illustrated in the figure (dotted lines). Draw the topology of the routing domain as it is “seen” by Router R2 (the numbers beside each link represent the utilization costs for the corresponding interfaces).

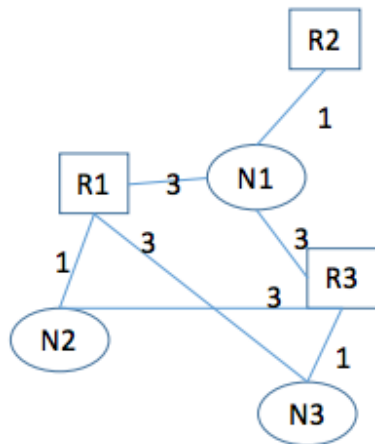


Solution

The OSPF protocol works according to the following two principles: (i) each router maintains a detailed representation of all links present in its own area, (ii) each router maintains a *synthetic* representation of all destinations (networks) reachable outside its own area.

In this specific case, Router R2 has a full, detailed vision of all links in Area 1 (to which it belongs), and he will “see” all networks external to Area 1 (N2 and N3) as reachable by passing through either router R1 and R3.

Here is the solution:



Exercise 3

The following table represents the connectivity graph of an IP network with 5 nodes over which a Link State routing protocol is used.

Da	Verso	Link	Costo	Sequence Number
A	B	1	2	2
A	D	3	2	2
B	A	1	2	2
B	C	2	2	2
B	E	4	2	2
C	B	2	2	2
C	E	5	2	2
D	A	3	2	2
D	E	6	2	2
E	B	4	2	2
E	C	5	2	2

For each of the following Link State Update messages sent by A, write how the network topology database changes:

From A, To B, Link 1, Cost=1, Sequence Number=1

From A, To B, Link 1, Cost=3, Sequence Number =3

From A, To F, Link 7, Cost=1, Sequence Number=1

Solution

From A, To B, Link 1, Cost=1, Sequence Number=1

This Link State Update (LSU) message has a sequence number=1, which is lower than the information stored about link A-B in the data base of the router. The message is hence discarded, and no change is performed on the data base.

Eventually the router sends to the source of the LSU message (i.e., the router who sent such message) the updated version of the link state (the one recorded, with Sequence Number =2, in the database)

From A, To B, Link 1, Cost=3, Sequence Number =3

The Link State Update message has a sequence number greater than the one corresponding to link A-B memorized inside the database of the router. The first line of the data base hence becomes:

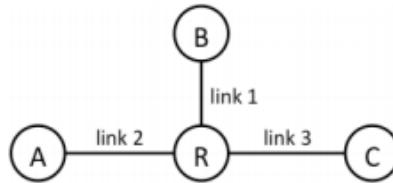
From A, To: B, link: 1, cost:3, Sequence Number:3

From A, To F, Link 7, Cost=1, Sequence Number=1

Link 7 (A-F) is not known by the router. The router therefore adds a line in its database:

From: A, To: F, link:7, cost: 1, Sequence Number: 1

Exercise 4:



In the above figure we have four routers, A, B, C e R, and the cost of using each of the three links is the same, and equal to 2. The reachability information of Router R is he following:

Destination	Cost	Next Hop
Net A	8	B
Net B	6	B
Net C	5	A
Net D	2	C

Router R receives from link 1 the following DV message: (NetA,4), (NetB,7), (NetC,4) and successively from link 2 the following DV message: (NetB,2), (NetC,3), (NetE,7).

Please write:

- the reachability information of R after the first DV message
- the reachability information of R after the second DV message
- After having received the second DV message, write the content of the DV message sent by router R, using Split Horizon with Poisonous Reverse.

Solution :

After the first DV:

Destination	Cost	Next Hop
Net A	6	B
Net B	9	B
Net C	5	A
Net D	2	C

After the second DV:

Destination	Cost	Next Hop
Net A	6	B
Net B	4	A
Net C	5	A
Net D	2	C
Net E	9	A

DV sent to A: (NetA,6),(NetB,infty),(NetC,infty),(NetD,2),(NetE,infty)

DV sent to B: (NetA,infty),(NetB,4),(NetC,5),(NetD,2),(NetE,9)

DV sent to C: (NetA,6),(NetB,4),(NetC,5),(NetD,infty),(NetE,9)